

Shreya Nakum

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Education

University of California, Irvine

June 2027

Bachelor of Science in Computer Science with Honors (GPA: 3.82 / 4.00)

Irvine, CA

- **Honors/Awards:** 1st Place Best Collaborative Research at UCI UROP 2025, Chancellor's Award for Excellence in Undergraduate Research, HackMIT Modal Honorable Mention, 6x Dean's Honors List.
- **Funding:** Calit2 Award 2026, UROP Award 2026, UROP Award 2025.
- **Involvements:** Information and Computer Science Honors Program, Campus Honors Collegium, Artificial Intelligence@UCI

Technical Skills

Languages: Python, C++, C, R, Javascript/Typescript, Java, SQL, MIPS (Assembly), MATLAB, Fortran 77, Fortran 90.

Frameworks & Libraries: PyTorch, TensorFlow/Keras, Flask, FastAPI, BeautifulSoup, scikit-learn, transformers, Robot Operating System (ROS).

Tools & Platforms: AWS (Lambda, Boto3), Firebase, Docker, PostgreSQL, Milvus, HuggingFace, Git, WordPress, Simulink, Linux.

Expertise: Computer vision, CNNs, RAG, Data Analysis, Statistical Modeling, Physical Simulations, Academic Writing, Spatial Rotation.

Other: Web Development, API Integration, Web Scraping, MLOps, Supercomputing.

Experience

NASA: Johnson Space Center

Jan 2026 – Aug 2026

AI/ML Engineering Intern

Houston, Texas

- Implemented **reaction wheel actuator models** and developed an **attitude control system** with **MATLAB/Simulink**.
- Implemented MATLAB/Simulink models in a high-fidelity **attitude dynamics simulation** framework with **Fortran 77** for commercial partners' LEO space stations utilizing non-propulsive attitude control.
- Developed computer vision algorithm to detect green & red lasers in a dissimilar altimeter for the terminal descent of lander modules using ROS and OpenCV.

UC Irvine: Donald Bren School of Information & Computer Science

Dec 2025 – Present

Researcher (Advisor: Dr. Krone-Martins)

Irvine, CA

- Tested the OpenAI and Gemini embedding models against multiple datasets about mathematics and physics to identify where the model lacks in training data through graphical evaluation with PacMAP.
- Compared the models' manifolds using the Davies-Bouldin index, mean silhouette, and intra-cosine similarity to determine their proclivity.

National Science Foundation: University of Oulu - M3S Lab

June 2025 – Aug 2025

Researcher (Advisor: Dr. Matteo Esposito)

Oulu, North Ostrobothnia, Finland

- Built a RAG pipeline to measure code clone similarity in software repositories.
- **Benchmarked 3 LLMs** across zero-shot, few-shot, and chain-of-thought prompts with 3k+ thresholding values.
- Automated statistical evaluation of model metrics; co-authoring a research paper.

UC Irvine: Rose Labs (School of Biological Sciences)

Sep 2024 – Present

Researcher (Advisor: Dr. Michael Rose)

Irvine, CA

- Trained 220 supervised ML models to identify evolving DNA sequences, achieving **80% accuracy**.
- Developed a multi-camera tracking algorithm implementing **2D pose estimation and identity propagation** for dense biological assays.
- Deployed a data annotation (Flask+JS) application that **accelerated data collection by 10x**.
- Designed a CNN object detection pipeline & trained a model (95% accuracy) to count *Drosophila melanogaster* eggs.
- Awarded 2 research prizes for novel small-object detection methods.

California Institute of Technology: Theoretical AstroPhysics Including Relativity and Cosmology

Oct 2021 – Mar 2023

Researcher (Advisor: Dr. Iryna Butsky)

Pasadena, CA

- Processed Hubble COS-Halos datasets with Python (**astropy, numpy, pandas**).
- Derived galaxy halo mass and velocity profiles; visualized results with **matplotlib**.
- Published findings in *MNRAS* and presented it at the American Astronomical Society conference.

Publications

Kemman, J., Nakum, S., Esposito, M., Cerny, T., Lenarduzzi, V., & Taibi, D. (2026). An LLM-based Technique for Detection of Code Clones Across Open Source Microservice Repository Forks. *Automated Software Engineering Journal*, **In Review**. <https://doi.org/10.13140/RG.2.2.34393.28004>

Butsky, I., Nakum, S., Ponnada, S., Hummels, C., Ji, S., & Hopkins, P. (2022). Constraining Cosmic-ray Transport with Observations of the Circumgalactic Medium. *Monthly Notices of the Royal Astronomical Society*, **521**(2), 2477–2483. <https://doi.org/10.1093/mnras/stad671>

Posters

Nakum, S., & Qi, M. (2025). Drift and Anti-Drift: Hundreds of Generations of Evolutionary Stasis. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California

Nakum, S. (2026). Detecting LLM Knowledge Gaps. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California

Black, A., Nakum, S., Gooty, N., & Greenspan, Z. (2026). Drosophila Neuroassay. Presented at the UC Irvine Undergraduate Research Symposium, Irvine, California

Nakum, S., Butsky, I., Hummels, C., Suoqing, J., Hopkins, P. (2022). Constraining the Cosmic-Ray Diffusion Coefficient with CGM Observations. Presented at American Astronomical Society Conference #240, Pasadena, California

Projects

HackMIT: ActsAI | *Honorable Mention, Modal.AI*

[GitHub](#)

- Built a mental health platform with a Llama-based chatbot to connect users to therapists; integrated SMS alerts for crisis detection.

Atoma Media: Financial News Sentiment Analysis

[HuggingFace](#)

- Built and automated Twitter scraper with PostgreSQL & AWS Lambda to collect financial news headlines and comments.
- Fine-tuned a sentiment-analysis model on financial news to predict market sentiment, achieving **80% accuracy** and **6k+ downloads**.